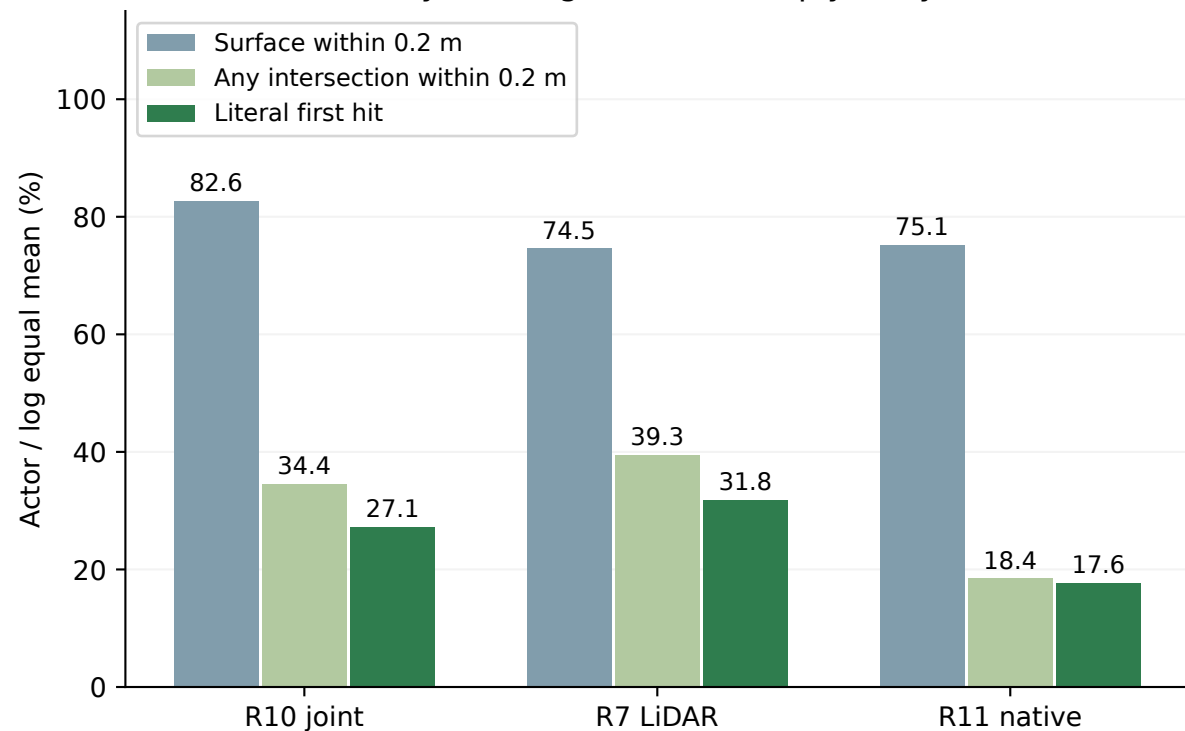
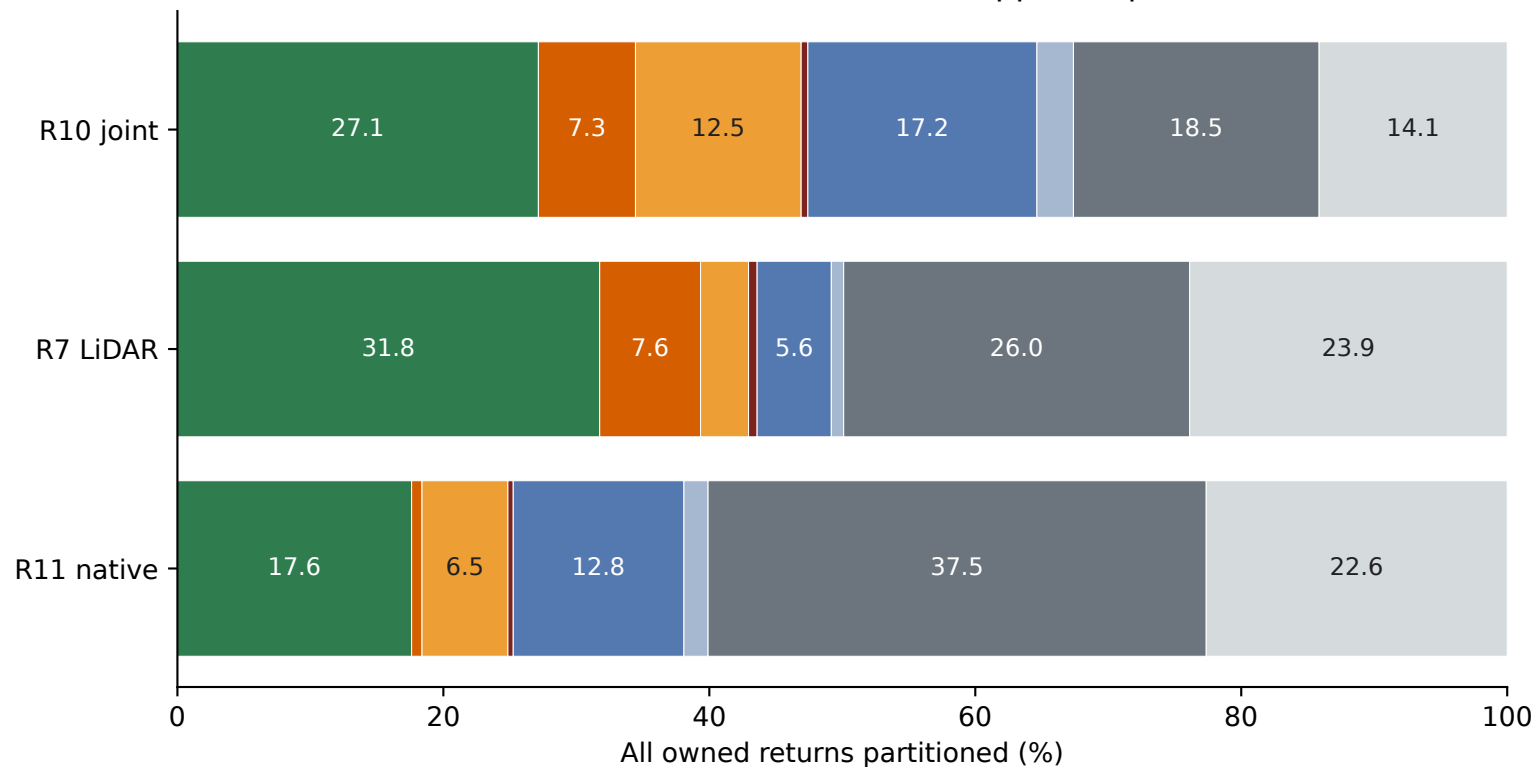


V7.3 fixed surfaces: coverage, support and first intersections

Nearby coverage does not imply a ray hit



First-return outcomes, with later support separated



- Correct first hit
- Early; later correct hit exists
- Early; no correct hit, surface nearby
- Early; no correct hit or nearby surface
- Late; surface nearby
- Late; no nearby surface
- Missing; surface nearby
- Missing; no nearby surface

75 development Actors / 5 logs; 11,886 owned ray occurrences; 23 Actors without owned returns retained. Shared full-track labels and hard-free semantics; model paths and optimizer costs differ. R11 counts reused.